Guide for AstroImageJ

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Steps to use AstroImageJ to extract light curve

1. Import file

   Select file → import → Image sequence... selecting the folder of the calibrated image. For mac users, this step would take a long time.

2. Find the target star in the image

   The target star is WASP11. You may search it on SIMBAD http://simbad.u-strasbg.fr/simbad and then compare the sky image provided by SIMBAD with our image.

3. Alignment

   Select Process → Align stack using WCS or apertures. Left click once to choose WASP 11 as the target star and Right click once to start the auto stacking process.

4. Photometry

   Select Analyze → multi-aperture. You can set the radius of the aperature used for photometry. You can save the results in the panel shown in Fig. 1. Save the results to

   ./input/light_curve/light_curve.csv.

   The relative flux is rel_flux_T1; the error of it is rel_flux_err_T1; the time of observation is JD_UTC.

   You may try to make the photometry error as small as you can (maybe you need to rerun the photometry pipeline). The methods include: select the used reference stars, select reference stars manually, change the radius of the used aperture. Why could you achieve a better (or not) result by doing so?

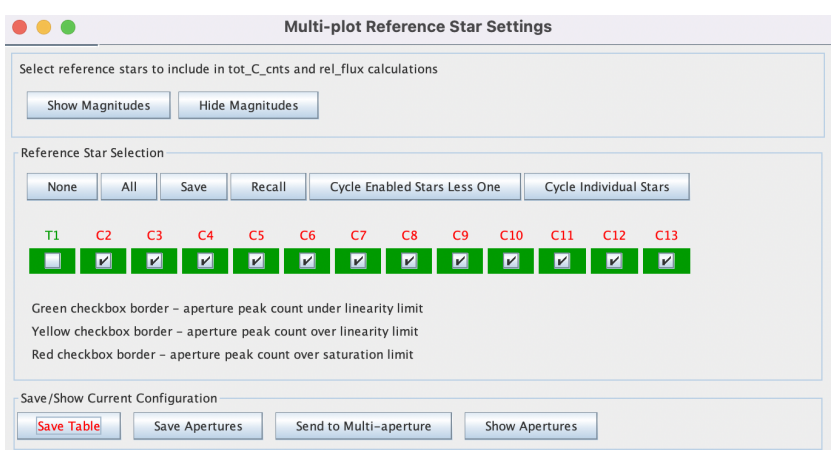

Figure 1: Panel for saving the photometry result